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Bhuvann Vinodh Ram

Data and AI Graduate · MSc AI for Medicine (UCD)

Stamp 1G holder: eligible to work full-time in Ireland without a work permit

PROFILE

MSc AI for Medicine graduate (UCD) working across the data lifecycle: automated pipelines and SQL data modelling with data-quality tests; statistics, A/B testing and dashboards; and training, explaining and deploying machine learning for imaging, signals, text and genomics. Every project below has public code, most with tests, CI and a live demo. Currently a Trust and Safety Associate at Accenture, Dublin.

EDUCATION

MSc Artificial Intelligence for Medicine and Medical Research — **2024 – 2025**
University College Dublin — GPA 3.37 / 4.2

BTech Computer Science and Medical Engineering — **2020 – 2024**
Sri Ramachandra Institute of Higher Education and Research, Chennai — CGPA 3.18 / 4

TECHNICAL SKILLS

Languages	Python, SQL, R, MATLAB (basic), JavaScript, Bash
Machine learning	scikit-learn, XGBoost, TensorFlow/Keras, PyTorch; CNNs, BiLSTM, VAEs, U-Net; SHAP
Data	pandas, NumPy, SciPy, SQL data modelling (dbt-style layers, data-quality tests), ETL pipelines, REST APIs, web scraping
Statistics	A/B testing (SRM, CUPED, power and MDE, multiple-testing correction), hypothesis testing, survival analysis
Vision	OpenCV, OpenCV.js, Detectron2, Tesseract.js
Visualisation	Tableau, matplotlib, seaborn, Plotly
Engineering	Git, GitHub Actions, GitLab CI/CD, Docker, pytest, Streamlit, Gradio, MySQL, MongoDB (basic)

EXPERIENCE

Trust and Safety Associate — **Mar 2026 – Present**
Accenture — Dublin, Ireland

- Evaluate, classify and moderate user-generated content against client platform policies to keep online communities safe and compliant.

Research Intern — **Apr 2025 – Sep 2025**
University College Dublin — Dublin, Ireland

- Integrated TCGA colon adenocarcinoma RNA-seq, copy-number and DNA methylation data and trained a variational autoencoder to learn a shared latent representation.
- Clustered the latent space (K-Means, HDBSCAN), evaluated with silhouette scores, and tested clinical relevance with Kaplan–Meier curves, log-rank tests, SHAP and STRING networks.

Project Intern — **Jan 2024 – Jun 2024**
Centre for Toxicology and Development, SRIHER — Chennai, India

- Built a video-to-data pipeline for zebrafish behaviour analysis used in drug testing: tracked fish across tanks and layers, computed speed and velocity, and produced charts and reports.

Research Intern — **Dec 2022 – Feb 2023**
IIITDM Kancheepuram — Chennai, India

- Developed and compared CNN breast cancer classifiers using pre-trained VGG16, ResNet, MobileNet and EfficientNet models.

Intern Feb 2022 – Apr 2022
Smartail Pvt. Ltd. Chennai, India

- Containerised web applications with Docker and deployed them through GitLab CI/CD pipelines.
- Worked on image processing with Detectron2, including hyperparameter tuning.

PROJECTS

Conflict Risk Monitor 2026
Python · pandas · scikit-learn · GitHub Actions · JavaScript Live demo · GitHub

- End-to-end pipeline: downloads UCDP conflict-event data (final and monthly releases) and World Bank indicators, builds a 176-country monthly panel, and publishes JSON to a static dashboard.
- Estimates the probability of 25+ battle deaths per country in the next three months; gradient boosting chosen on a validation period scored ROC AUC 0.989 and average precision 0.959 on unseen 2022–26 data.

Checkout redesign A/B test (SQL data modelling and experiment analysis) 2026
SQL · dbt-style models · Python · statistics · GitHub Actions GitHub

- Modelled messy raw event exports (simulated, 200,000 users) in nine SQL models across staging, intermediate and mart layers, with dbt-style data-quality tests.
- Experiment readout with SRM check, z-test, Welch t-test, bootstrap CI, CUPED, Holm correction and power/MDE: +5.2% relative conversion lift (p = 0.016).

Tamil Thedal (தமிழ் தேடல்) 2026
JavaScript · OpenCV.js · Tesseract.js · Python ETL Live demo · GitHub

- ETL script builds 1,549 verses from open sources at a pinned commit, keeps only public-domain fields, adds themes, and is rebuilt by GitHub Actions with tests.
- Search engine for Tamil literature (1,549 verses) with sound-alike matching of romanised Tamil and a TF-IDF chat helper that runs fully in the browser.

Ovarian cancer survival prediction (HGSOC) 2025
XGBoost · SHAP · TCGA GitHub

- XGBoost on TCGA clinical and mutation data with SHAP explanations; found and removed follow-up variables that leaked the outcome, then reran the notebook end to end in GitHub Actions: test accuracy 0.78, ROC AUC 0.91.

ECG heartbeat classification 2025
1D CNN · Keras GitHub

- 1D CNN classifying MIT-BIH heartbeats into five classes: 96.01% test accuracy on 21,892 held-out beats, reproduced by an automated GitHub Actions rerun.

OTHER PROJECTS

- **Brain tumour classification from MRI** (CNN · OpenCV · Streamlit · pytest, 2025): Four-class CNN (0.80 accuracy, 0.77 macro F1 on 2,258 test images) and ResNet50 transfer learning, deployed as a Streamlit app with unit tests and CI.
- **Emotion classification from tweets** (BiLSTM · TensorFlow · Gradio, 2025): Dual-input BiLSTM combining word embeddings with sentiment polarity: 90.63% test accuracy, served as a Gradio app.
- **Handwritten prescription reader** (Detectron2 · object detection, 2023): Detectron2 model that locates handwritten medicine names in prescription images, trained on a custom annotated dataset.
- **Student engagement analysis (VLE)** (scikit-learn · ensembles, 2025): Voting ensembles (random forest, logistic regression, gradient boosting, SVM) on virtual learning environment data, compared across feature-selection settings.
- **Retail sales register analysis** (Requests · Beautiful Soup · pandas, 2025): Scraped multi-page sales records, cleaned dates, currencies and missing values, and analysed revenue, profit and customer behaviour by location and age group.
- **RNA-seq analysis of HER2 breast cancer** (R · DESeq2 · clusterProfiler, 2024): Differential expression, pathway enrichment and survival predictors in R to study how HER2 amplification changes gene expression.

AWARDS AND ACTIVITIES

UCD Innovation Challenge 2024: Innovation Spark Award · NASA Space Apps Challenge 2024: built a climate storytelling website with a team · UCD Advantage Award · First place, Tamil poetry competition (Agal Tamil Club)